Guilherme Iglesias Rocha Gomes

Mechanical Engineering | Data & Software Engineering

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PROFILE

Final-year Mechanical Engineering student at IPB with hands-on experience in Python programming, data analysis and management, mathematical modelling, and computer vision systems. I have been involved in scientific research since 2020 — including a scientific initiation scholarship in Brazil and a final-year project in agricultural robotics with AI (YOLOv12n). I have experience creating structured datasets, documenting them using the Croissant/JSON-LD format, computing SHA-256 checksums for data integrity, and familiarity with property graph data models (MoMa/PG-JSON/Neo4j). Career goal: Software Engineer or Data Engineer.

TECHNICAL SKILLS

Programming & Data

- Python — analysis, CV, ML
- C# — Unity, systems
- Java — front-end
- SQL — relational databases
- HTML / CSS
- JSON / JSON-LD / Croissant

Engineering & Design

- SolidWorks / AutoCAD
- Autodesk Inventor
- ANSYS & Fluent Meshing
- Matlab / Simulink
- 3D Printing and CNC
- Arduino

AI & Data Management

- OpenCV — computer vision
- YOLOv12n / Ultralytics
- Mathematical modelling
- Dataset creation
- Graphs MoMa / Neo4j
- SHA-256 / data integrity

Tools & Standards

- Microsoft Excel
- Unity 3D Engine
- Croissant spec v1.0
- Git / GitHub
- PG-JSON / Neo4j

EDUCATION

Bachelor's Degree in Mechanical Engineering — IPB — Polytechnic Institute of Bragança *Sep. 2023 – Present*

- Final-year student (EQF Level 6) — Bragança, Portugal
- **Final-year project:** Stereo Vision System for Vineyard Robotics using dual ESP32-CAM, Python, OpenCV and YOLOv12n
- Presented at IN2COP 2026 International Conference, Bragança Polytechnic University, Jan. 2026

Mechanical Engineering Technician — Integrated Secondary Education — IFES — Federal Institute, Vitória, Brazil *Feb. 2019 – Dec. 2022*

- Integrated technical and scientific training (EQF Level 4)
- Foundation in mechanics, automation, programming and industrial projects

WORK EXPERIENCE

Mechanical Engineering Intern — Cut Out — Bragança, Portugal *Jun. 2025 – Dec. 2025*

- Production support in 3D printing, laser cutting and CNC, from preparation to quality control
- 2D and 3D modelling with SolidWorks and AutoCAD
- Process optimisation and team workspace organisation

Private Tutor — Universo do Saber — Bragança, Portugal *Dec. 2023 – Mar. 2025*

- Subjects: Mathematics, Calculus, Algebra and English — 150 teaching hours
- Students ranging from primary school to university level

RESEARCH PROJECTS

Final-Year Project — Stereo Vision for Agricultural Robotics — IPB *Nov. 2025 – Present*

- Low-cost system with dual ESP32-CAM, Python, OpenCV and YOLOv12n for autonomous navigation in vineyards
- Camera calibration, hardware synchronisation, real-time depth mapping

- Presented at IN2COP 2026 International Conference, Bragança

Erasmus — University Registration System — Vilniaus Kolegija, Lithuania *Nov. 2025*

- Registration system using SQL, HTML/CSS and Java; relational database design and management

Erasmus — Robotic Arm — Vilniaus Kolegija, Lithuania *May 2025*

- Mechanical design in SolidWorks, 3D prototyping, programming with Arduino and Python

Erasmus — Digital Challenge Innovation Lab (D-Chill) — Hanze University, Groningen, Netherlands *Dec. 2024*

- Educational game to teach Python to children, developed in partnership with SharkCoders
- Participation in IN2COP 2025 International Conference, IPB

Research Grant — BiWit / DataGEMS Platform — IPB — L2I, Bragança *Apr. 2026 – Present*

- Development of mathematical models and data management systems within the DataGEMS platform (NORTE2030/FEDER)
- Work with Croissant/JSON-LD dataset documentation standard and MoMa property graph model
- Self-directed study of Croissant spec v1.0, PG-JSON format, SHA-256 integrity verification and Neo4j graph databases

Scientific Initiation Scholarship — Industrial Virtual Reality — Vale / IFES — Vitória, Brazil *Nov. 2020 – Mar. 2023*

- VR training simulator for industrial machine inspection (IFES/Vale partnership)
- Serious game in Unity3D with C# and Python; creation of AI training datasets
- Scientific articles presented at SVR 2021 and SVR 2022 symposia
- Instructor in serious game development courses

LANGUAGES

Portuguese: Native language **English:** C2 Proficient (listening, reading, spoken production and interaction, writing)